

ALBERT HO

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EDUCATION

Bachelor of Science, Computer Science | University of California, San Diego
Awards: Summer URS Philip and Elizabeth Hiestand Scholarship

June 2026
GPA: 3.7

EXPERIENCES

Software Engineer Research Intern | UCSD Health (Hojun Li Lab)

October 2024 - Present

- Engineered a scalable Python-based bioinformatics pipeline for parsing BAM alignment files from Oxford Nanopore's Dorado platform, enabling detection and quantification of RNA editing events
- Developed a computationally optimized algorithm to calculate per-base modification frequencies, leveraging ONT's statistical models and vectorized Pandas/Numpy operations to minimize runtime and memory overhead
- Implemented automated genomic feature annotation using interval tree data structures, producing reproducible datasets that support downstream analyses
- Engineered a reproducible, scalable workflow for more than 57,000 single-cell profiles using R, Seurat, and Python/Pandas, improving pipeline maintainability and computational efficiency

Frontend Engineer Intern | Cadent

June 2026 - August 2026

- Engineered an AI-assisted software development workflow that transforms Jira Epics and Stories into structured, implementation-ready plans, executes those plans through Cursor agents, and generates and pushes pull requests
- Built foundational skills, references, and infrastructure for expansion of the company's AI-assisted development ecosystem
- Integrated Atlassian and Storybook MCPs to provide context to project requirements and documentation in the workflow
- Designed guidelines and standardized references for Cursor skills, agents, and subagents ensuring consistent outputs

Data Science Intern | Lawrence Livermore National Laboratory | [Link](#)

July 2024 - August 2024

- Developed a production-ready machine learning pipeline for ventricular arrhythmia classification, benchmarking Logistic Regression, Random Forest, and deep learning models (CNNs and RNNs)
- Achieved 99% recall by designing and hyperparameter-tuning a sequence-to-sequence convolutional neural network, significantly improving diagnostic sensitivity
- Processed and labeled large datasets (over 16,000 samples) using Pandas dataframes for training and testing

PROJECTS

kumQAt (AI QA Automation Platform) | TypeScript, Next.js, FastAPI, Python, Gemini | [Link](#)

April 2026

- Led product and workflow design for an AI-powered QA platform converting natural language into executable testing pipelines
- Built a full-stack app with async orchestration enabling parallel execution and real-time streaming via Server-Sent Events
- Designed a multi-stage LLM pipeline (generation, execution, validation) producing structured results with triage-ready outputs
- Implemented autonomous browser-based testing with execution traces, screenshots, and persistent run history, supporting reruns and scheduled QA workflows

Hall Pass | TypeScript, React Native, Firebase, AWS S3, Expo | [Link](#)

January 2026 - Present

- Designed a full-stack social media platform for universities, managing the project lifecycle from UI design to deployment
- Implemented backend using Firebase for real-time messaging and AWS S3 for image storage, implementing private course-specific discussion spaces and verification
- Utilized AI-augmented development workflows to assist feature prototyping and optimize TypeScript codebases, while maintaining system logic and security protocols

TECHNICAL SKILLS

Languages | Python, R, C/C++, Swift, TypeScript, HTML, CSS, JavaScript, Java

Technologies | Agile, React, Node.js, Tailwind CSS, Git, Github, VS Code, Firebase, AWS, PyCharm, XCode, scikit-learn, tensorflow, PyTorch, pandas, NumPy, Matplotlib, D3.js

Relevant Coursework | Data Structures and Algorithms, Operating Systems, AI: Probabilistic Models, Software Engineering, Engineering Probability and Statistics, Intro to Cryptography, Theory of Computation

INTERESTS

Machine Learning, Bioinformatics, Cybersecurity (Cryptography & Exploit Development), Playing Guitar/Piano, Global Travel